Mkoba System Critique

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This is a critique of the Mkoba system (mkoba.digital) as it stands after the 12-item roadmap build. Weaknesses are listed most serious first. They come from building and testing the system; some are judgment rather than tested fact.

1. Too much was built before any real user

All 12 roadmap features exist, but none has met a real vendor or a real outside service. Selcom, Google, Africa's Talking and Meta were all simulated, so the 299 passing tests show the logic is consistent, not that payments, voice or USSD actually work. Every feature is something one person has to maintain and support.

A narrow launch would teach more: records, debts and reports for 20–50 vendors, measuring whether they keep using it.

2. WhatsApp's 24-hour rule breaks most notifications

Mkoba can only message someone who wrote to it within the last 24 hours, unless Meta has approved message templates (still an open item). So:

- a customer order can sit unseen until the vendor next sends a message;
- riders can miss jobs;
- bulk-order closing and payment messages can fail to arrive.

The order, delivery and bulk features depend on getting templates approved first.

3. The command language is fragile for the target users

- `nimelipa Juma 2000` is saved as an expense, while `nimemlipa Juma 2000` is a debt payment. One letter changes the accounting, and nothing warns the vendor.
- A small number without TSh is read as a count (`soda 10` means 10 sodas, not TSh 10).
- A two-word name saves under the last word.
- English is missing some commands, such as paying back a debt.

A message saved quietly under the wrong category is worse than "I didn't understand", because the vendor's numbers become wrong without them knowing.

4. Mkoba holding money is a legal and trust risk

All payments go into one Mkoba Selcom account, and the admin pays vendors and riders by hand. That probably makes Mkoba a payment intermediary under Tanzanian rules. It also brings reconciliation work, room for human error or fraud, and blame when a payout is late.

Check this with a lawyer before turning payments on.

5. The credit score looks more reliable than it is

- The weights (30/25/15/15/10/5) are reasoned guesses, not checked against real repayment data.
- Most inputs are typed by the vendor, so inflating sales raises the score. Only pasted mobile-money SMS are close to evidence.

Showing lenders a number from 0 to 100 suggests a precision it doesn't have. Call it an "activity score" until it's validated.

6. Data protection is unresolved

Under Tanzania's PDPA you likely need to register with the data protection commission. Voice notes go to Google, which is likely abroad, so check the rules on sending data outside Tanzania. The data location question in the manual is still open.

7. Costs per vendor may be higher than what vendors can pay

Each of these costs money: Meta conversations, Google voice transcription per minute, USSD sessions and the shortcode lease, and Selcom fees. The revenue model is not yet defined, and that is the gap. Without it, the more active a vendor is, the more they cost Mkoba.

8. Engineering debt

- **Four interfaces to maintain:** WhatsApp, dashboard, app and USSD. They are already inconsistent: the dashboard is Swahili-only while the app is bilingual.

- **Overloaded data model:** households are stored as vendors, and one phone number can hold only one role. A vendor can't also be a rider or a household on the same number.
- **Regex commands spread across modules:** about 15 modules each match their own keywords, and the order in which webhook.js tries them decides the result. Collisions will grow as commands are added; "order", for example, means different things to a customer and a vendor.
- **Duplicated text:** English and Swahili replies are written separately in each module, which is how gaps like the missing English debt command happen.
- **Offline records can be lost:** unsent records in the app live in the browser's storage and disappear if it's cleared.

9. Bulk buying is mostly operations, not software

The code collects what each vendor needs. It doesn't decide who the wholesaler is, who sets the price, who moves the goods, or what happens when someone doesn't pay their share. The Friday close is fixed in the code.

What's solid

- Records and summaries
- Debts
- Confirming before saving anything uncertain, including every voice note
- Payment status always re-checked with Selcom, never trusted from the webhook
- Privacy deletion flow
- Security basics: same-origin checks, scrypt passwords, timing-safe comparisons

Recommendation

1. Launch only the core first: records, debts and reports.
2. Sort out the Meta templates and the payment-licensing question before switching on orders, delivery, bulk buying and payments.
3. Fix the `nimelipa` trap: when the name matches someone in the debts list, Mkoba asks "a debt to Juma, or an expense?"